

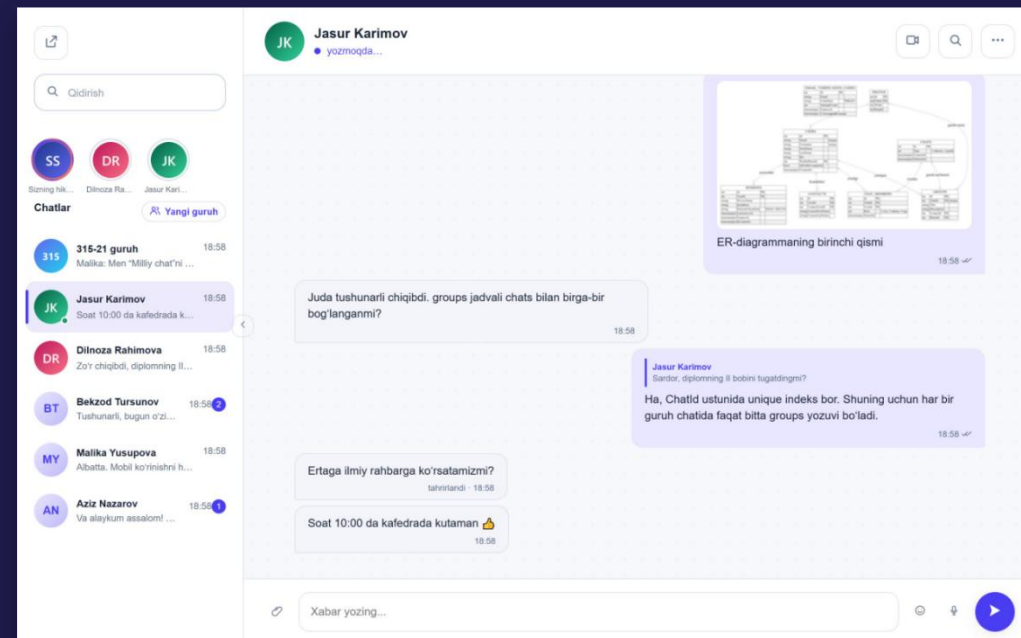
BITIRUV MALAKAVIY ISHI

“Milliy chat” dasturini ishlab chiqish

Telegramga o‘xshash milliy veb-messenjer

Bajardi: **S.O. Sohinazarov**

Ilmiy rahbar: _____



Mavzuning dolzarbligi



“Raqamli O‘zbekiston – 2030” strategiyasi

PF-6079-son Farmon: mahalliy dasturiy mahsulotlar va axborot xavfsizligi ustuvor vazifa



Ma’lumotlarni lokalizatsiya qilish talabi

“Shaxsga doir ma’lumotlar to’g’risida”gi Qonun: fuqarolar ma’lumotlari respublika hududida saqlanishi kerak



Xorijiy messenjerlarga bog‘liqlik

Telegram, WhatsApp, Signal, WeChat — barchasi xorijiy yurisdiksiyada, ma’lumotlar mamlakatdan tashqarida

Yechim — “Milliy chat”

- ✓ Ma’lumotlar O‘zbekistondagi serverda saqlanadi
- ✓ O‘zbek tilidagi tanish interfeys
- ✓ Kod va arxitektura mahalliy nazoratda
- ✓ Tashkilotlar uchun ichki aloqa vositasi

Ishning maqsadi va vazifalari



Maqsad: foydalanuvchi ma'lumotlarini o'z serverida saqlaydigan, shaxsiy va guruh yozishmalarini real vaqtda ta'minlaydigan "Milliy chat" veb-messenjerini loyihalash va ishlab chiqish.

01

Mavjud messenjerlar va real vaqt texnologiyalarini tahlil qilish

02

Tizimga funksional va nofunksional talablarni aniqlash

03

Arxitektura, ma'lumotlar bazasi va UML modellarini loyihalash

04

Xavfsiz autentifikatsiya va sessiyalar tizimini yaratish

05

Server (.NET 8) va klient (Angular) qismlarini ishlab chiqish

06

Bulutga joylashtirish, sinash va HFX talablarini ishlab chiqish

Mavjud messenjerlar tahlili

Mezon	Telegram	WhatsApp	Signal	WeChat	Milliy chat
Egasi (davlat)	BAA	Meta (AQSh)	AQSh	Tencent (Xitoy)	Mahalliy
Ro'yxatdan o'tish	Telefon raqami	Telefon raqami	Telefon raqami	Telefon raqami	E-pochta kodi, Google
E2E shifrlash	Faqat maxfiy chatlarda	Sukut bo'yicha	Sukut bo'yicha	Yo'q	Rejalashtirilgan (TLS bor)
Yozishmalar	Bulutda	Qurilmada	Qurilmada	Serverda	O'z serverida
Server kodi	Yopiq	Yopiq	Ochiq	Yopiq	Mahalliy nazoratda
Ma'lumotlar joyi	Xorijda	Xorijda	Xorijda	Xitoyda	O'zbekistonda ham mumkin



Barcha mashhur messenjerlar ma'lumotlarni xorijda saqlaydi. Foydalanish qulayligi bo'yicha namuna sifatida Telegram tanlandi.

Real vaqtda aloqa texnologiyalari

Texnologiya	Yoʻnalish	Kechikish	Brauzerlar
Short polling	Mijoz → server	Yuqori	Barchasi
Long polling	Mijoz → server	Oʻrtacha	Barchasi
Server-Sent Events	Server → mijoz	Past	Zamonaviy
WebSocket	Ikki tomonlama	Juda past	Zamonaviy
SignalR	Ikki tomonlama	Juda past	Barchasi (fallback)



Nega SignalR?

WebSocket



SSE



Long polling

transport avtomatik tanlanadi

- Hab (Hub) abstraksiyasi: server va mijoz bir-birining metodlarini chaqiradi
- Guruhlar, JWT autentifikatsiyasi va qayta ulanish tayyor holda
- .NET tarkibiy qismi; brauzer uchun rasmiy @microsoft/signalr

Tizimga qo'yilgan talablar

12

funksional talab

- Kirish: e-pochta kodi yoki Google
- Profil va faol sessiyalar
- Shaxsiy va guruh chatlari
- Javob, tahrirlash, o'chirish, qidiruv
- Rasm, video va fayllar
- "Yozmoqda", o'qildi, onlayn holat
- 24 soatlik hikoyalar
- Yorug' / qorong'i mavzu, mobil ko'rinish

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nofunksional talab

- Unumdorlik: xabar < 1 soniyada
- Xavfsizlik: HTTPS, xeshlar, huquqlar
- Ishonchlilik: avtomatik qayta ulanish
- Kengaytiriluvchanlik: holatsiz API
- Qo'llab-quvvatlanuvchanlik
- Ko'chiriluvchanlik: Docker
- Qulaylik: o'zbek tilidagi interfeys

Tizim aktorlari



Mehmon

kirish, ro'yxatdan o'tish



Foydalanuvchi

chatlar, xabarlar,
hikoyalar



Guruh egasi

a'zolar va rollarni
boshqarish

Texnologiyalar steki



Server qismi

- .NET 8, ASP.NET Core
- SignalR — real vaqt
- Entity Framework Core 8
- PostgreSQL 16
- FluentValidation, JWT
- Redis — onlayn holat



Klient qismi

- Angular 22 (standalone, signals)
- TypeScript, RxJS
- @microsoft/signalr
- Google Identity Services
- O'z dizayn tizimi (SCSS)
- Server tomonida render (SSR)



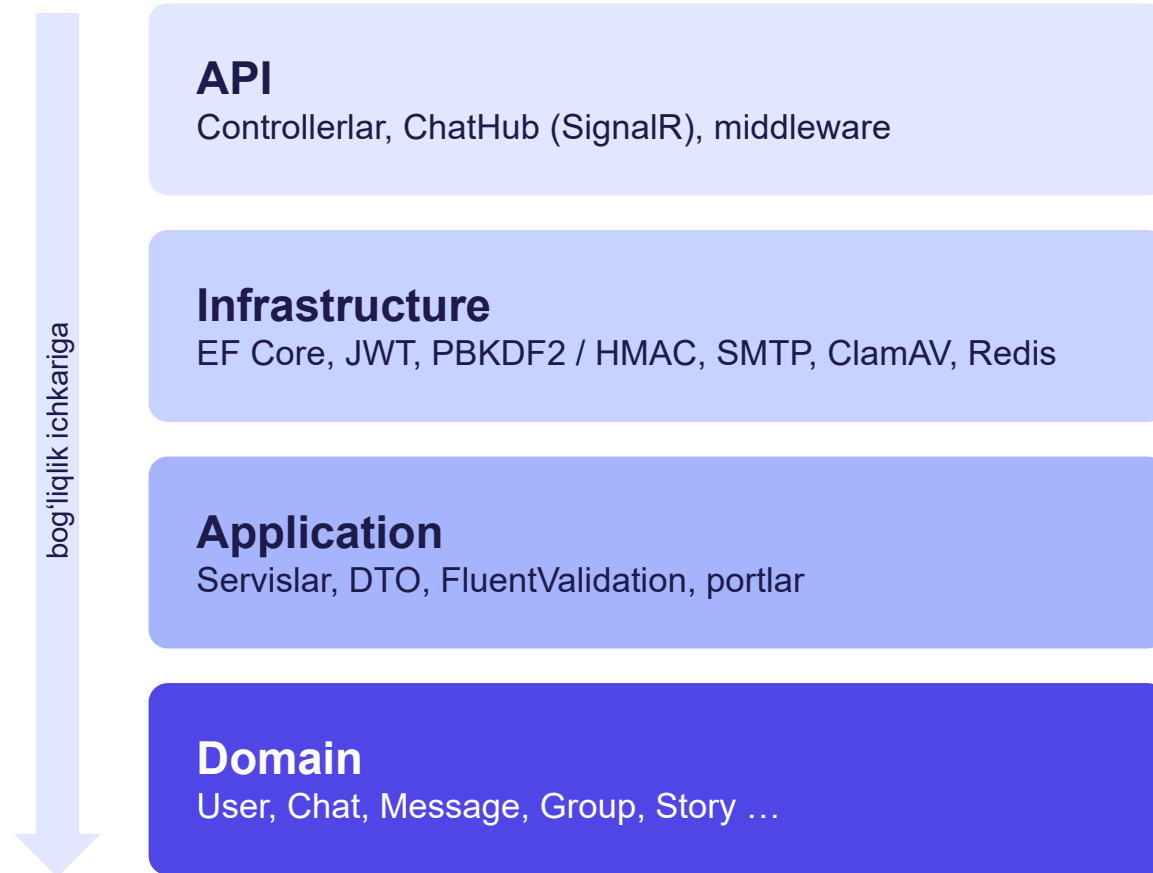
Infratuzilma

- Docker (multi-stage)
- Render bulut platformasi
- SMTP — kodlar yuborish
- ClamAV — antivirus
- ImageSharp, FFmpeg — media
- Git va GitHub

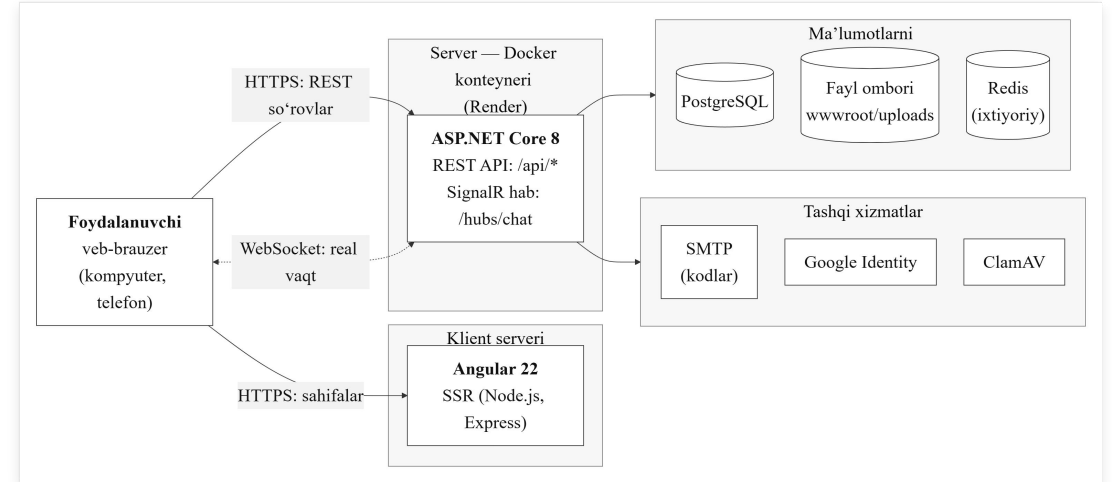
✓ Barcha texnologiyalar ochiq kodli va bepul — litsenziya xarajati yo'q.

Tizim arxitekturası

Clean Architecture — server qatlamlari



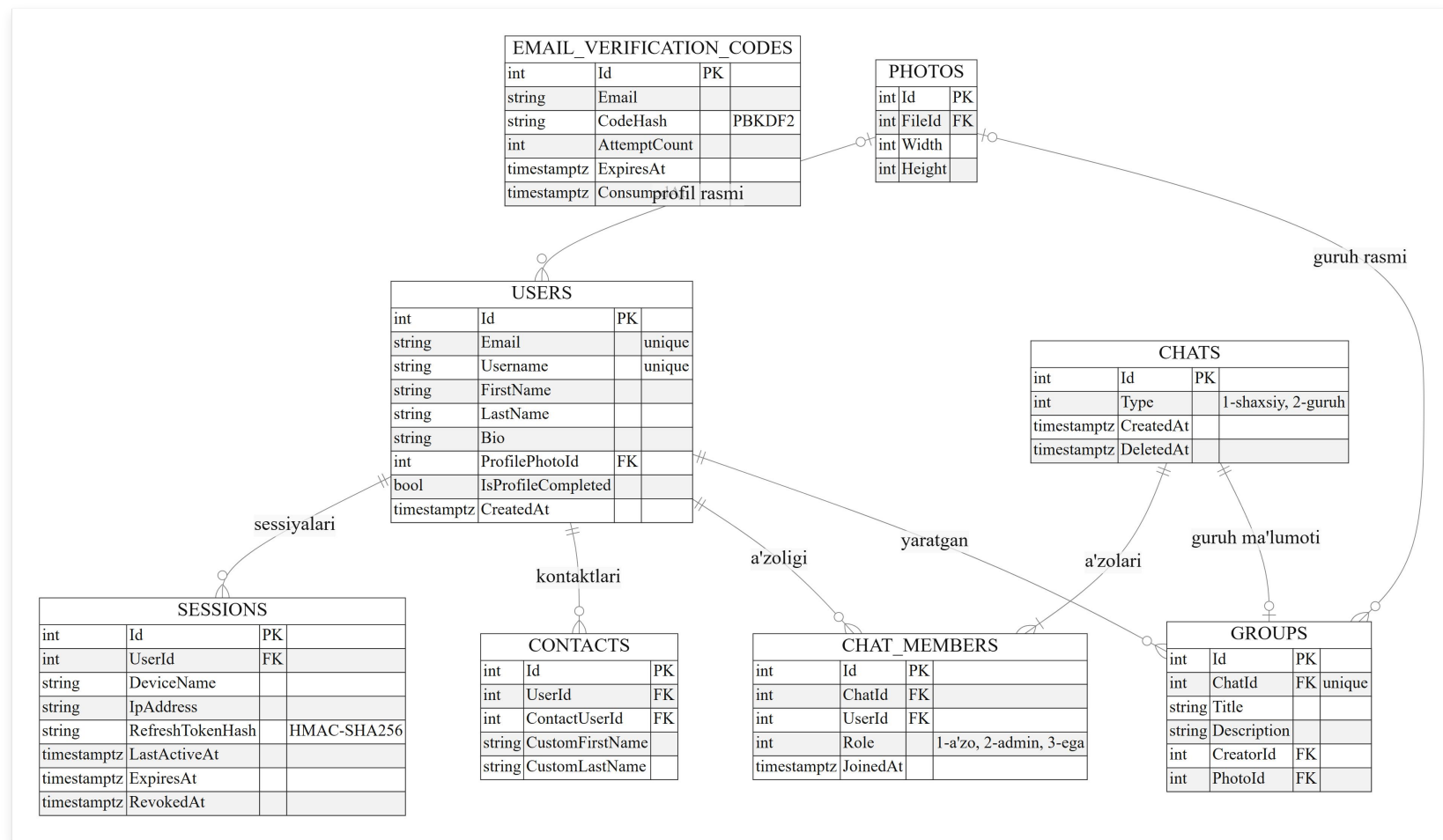
Joylashtirish diagrammasi



Klient (Angular) tuzilmasi

- features/* — har bir bo'lim alohida: auth, chat, groups, stories
- data-access — HTTP va SignalR servislari
- signals asosida holat, SSR bilan tez ochilish

Ma'lumotlar bazasini loyihalash



9

sxema

identity, messaging, groups, stories ...

32

jadval

kelajakdagi kanallar va botlar uchun ham

Keyset

kursorli sahifalash

beforeId + limit (1–100), totalCount siz

Xavfsizlik va autentifikatsiya

1

Elektron pochtaga 6 xonali kod

Bazada faqat PBKDF2 xeshi saqlanadi, parol umuman yo'q

2

Yoki Google hisobi orqali

Google ID tokeni serverda tekshiriladi

3

JWT access token — 15 daqiqa

Har bir API so'rovi va SignalR ulanishida tekshiriladi

4

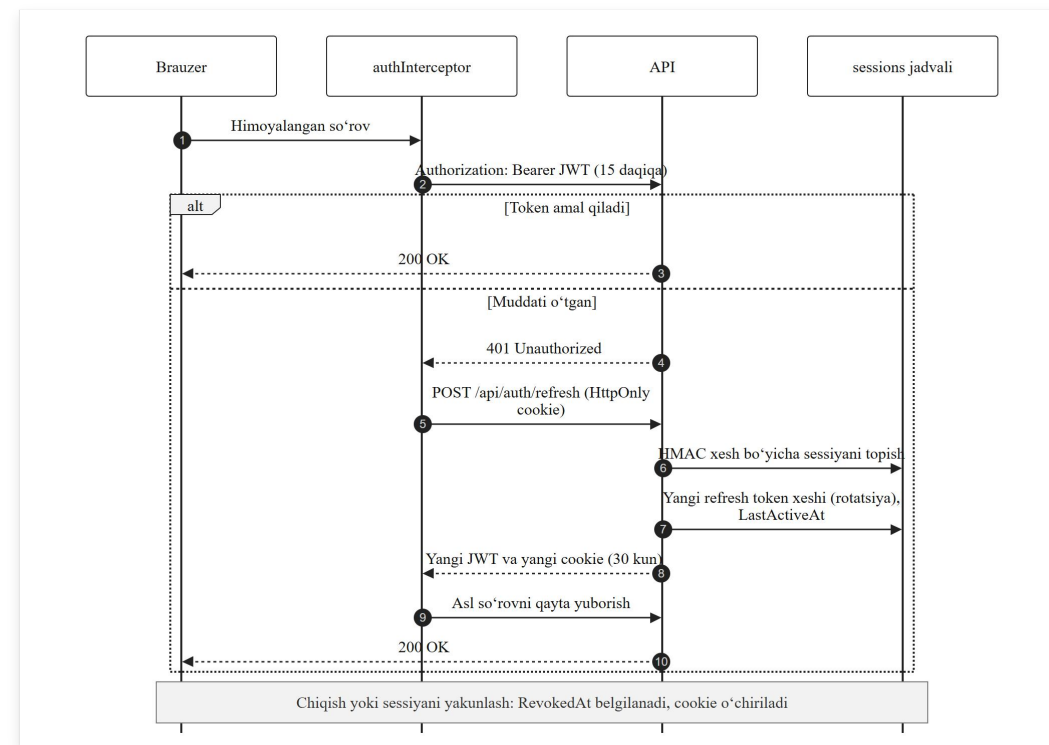
Refresh token — 30 kun, rotatsiya

HttpOnly cookie'da, bazada HMAC xeshi; har safar almashadi

5

Qurilmalar bo'yicha sessiyalar

Bitta yoki boshqa barcha sessiyalarni yakunlash mumkin

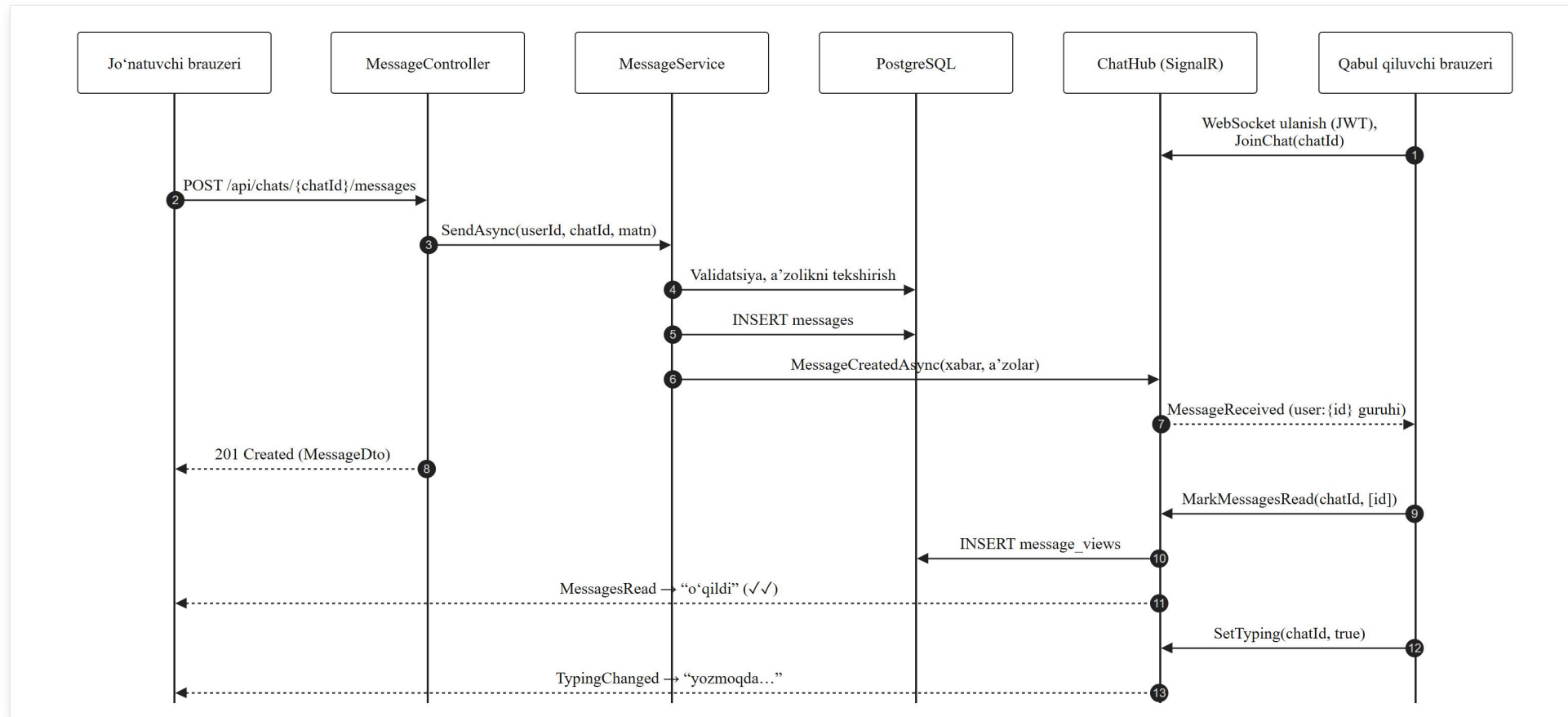


Har so'rovda chat a'zoli tekshiriladi



Fayllar ClamAV antivirusidan o'tadi

Real vaqtda xabar yuborish jarayoni



1

REST: xabar tekshirilib, bazaga yoziladi

2

ChatHub: chat a'zolariga darhol yuboriladi

3

O'qildi ✓✓ va "yozmoqda..." belgilari

Dastur interfeysi: tizimga kirish

1 MILLIY CHAT

Xush kelibsiz

Akkauntingiz yo'qmi? [Ro'yxatdan o'ting](#)

☐ Tasdiqlash kodi emailingizga yuboriladi.

Email manzilingiz

☐ Ushbu qurilmada eslab qolish [Yordam kerakmi?](#)

[Davom etish](#)

yoki davom eting

[Passkey bilan](#) [QR kodni skanerlash](#)

Davom etish orqali Foydalanish shartlari va Maxfiylik siyosatiga rozilik bildirasiz.

Kirish

E-pochta manzilini kiritish yoki Google orqali kirish

2 MILLIY CHAT

Kodni tasdiqlang

yangi.talaba@example.com manziliga yuborilgan 6 xonali kodni kiriting.

☐ Kod 10 daqiqa davomida amal qiladi.

Tasdiqlash kodi

[Kodni tekshirish](#)

[Email manzilini o'zgartirish](#)

Davom etish orqali Foydalanish shartlari va Maxfiylik siyosatiga rozilik bildirasiz.

Tasdiqlash kodi

6 xonali bir martalik kod pochtaga yuboriladi

3 MILLIY CHAT

Profil yarating

Siz uchun Milliy chatdagi profilingizni sozlaymiz.

☐ Ma'lumotlaringiz faqat profilingizda ko'rsatiladi.

Ism Familiya

Username

5-50 belgi: harf, raqam va pastki chiziq ishlatiladi.

[Profilni yaratish](#)

[Email manzilini o'zgartirish](#)

Davom etish orqali Foydalanish shartlari va Maxfiylik siyosatiga rozilik bildirasiz.

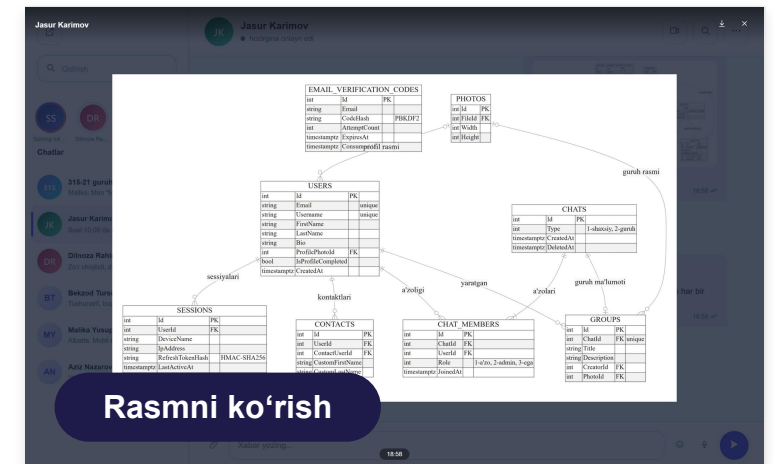
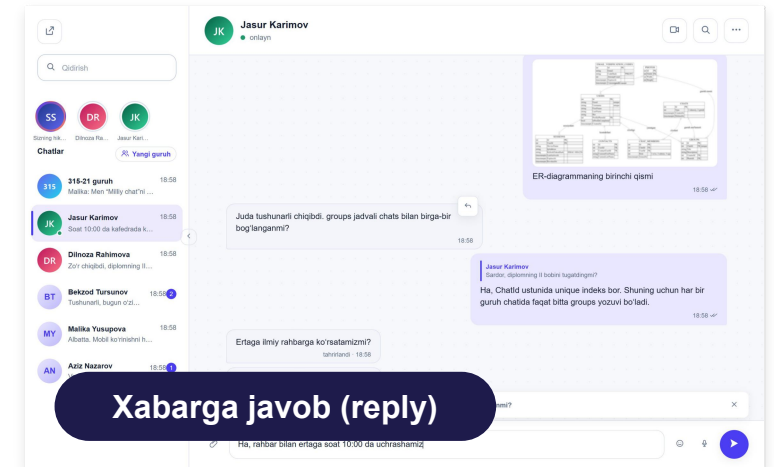
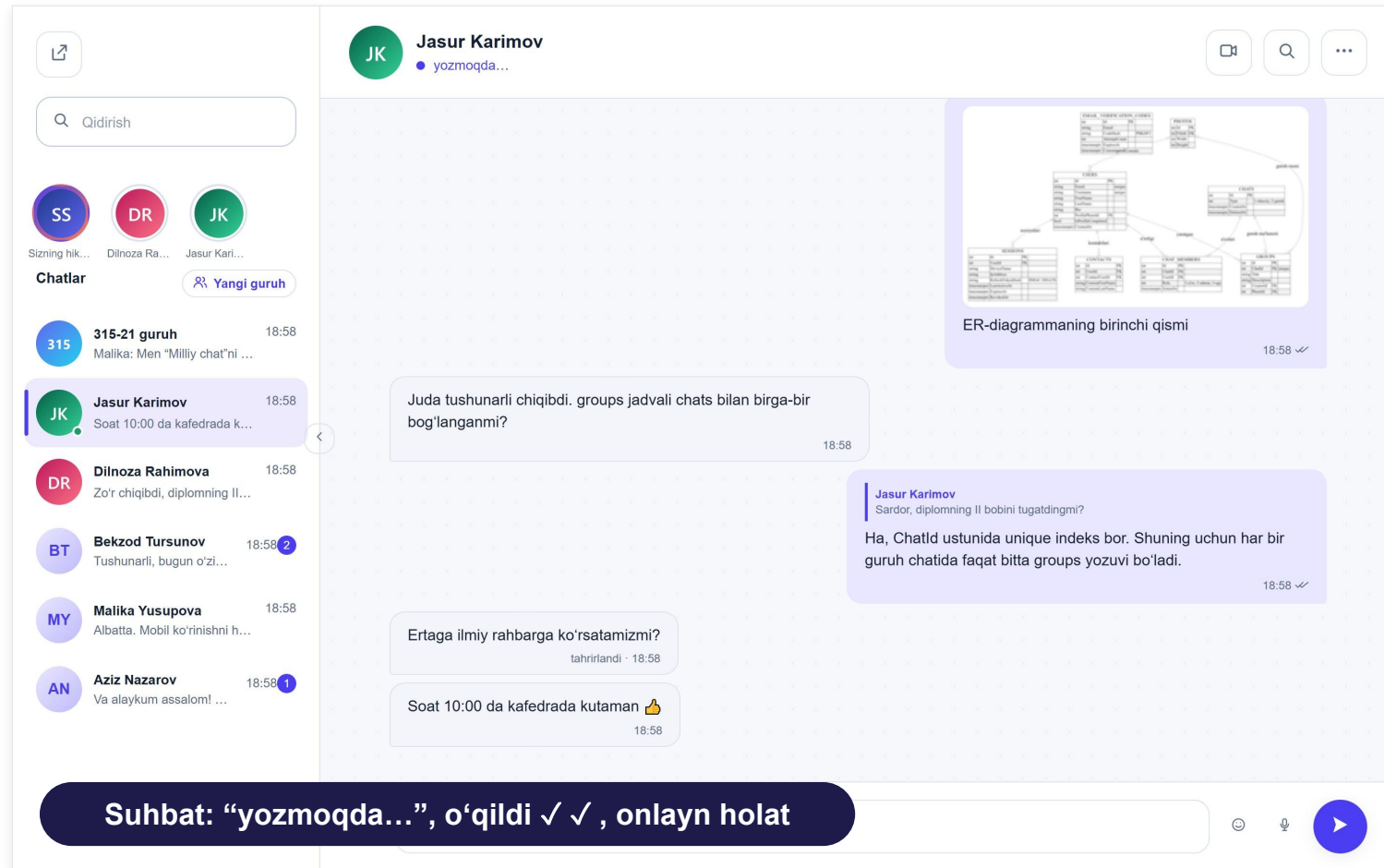
Ro'yxatdan o'tish

Yangi foydalanuvchi ismi va username'ini kiritadi



Parolsiz kirish: parol saqlanmaydi va sizib chiqmaydi — foydalanuvchi faqat pochtaga egaligini tasdiqlaydi.

Dastur interfeysi: shaxsiy chat



Dastur interfeysi: guruh chatlari



Egasi

Adminlarni tayinlaydi, guruhni o'chiradi



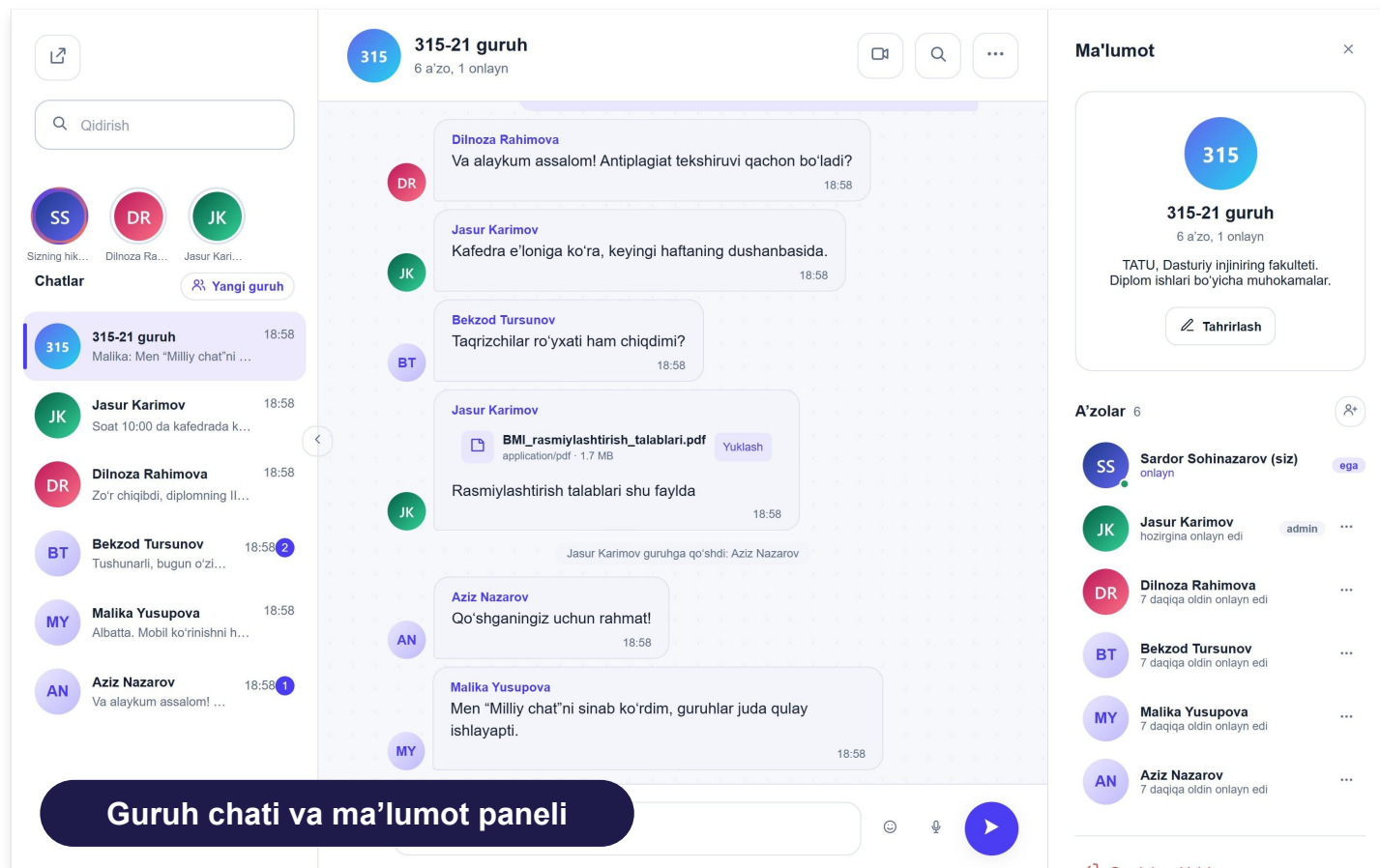
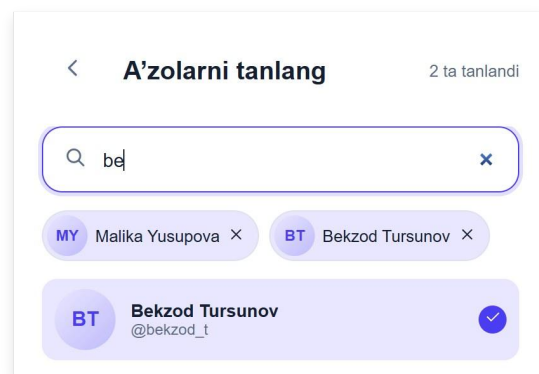
Administrator

A'zo qo'shadi va chiqaradi, guruhni tahrirlaydi



A'zo

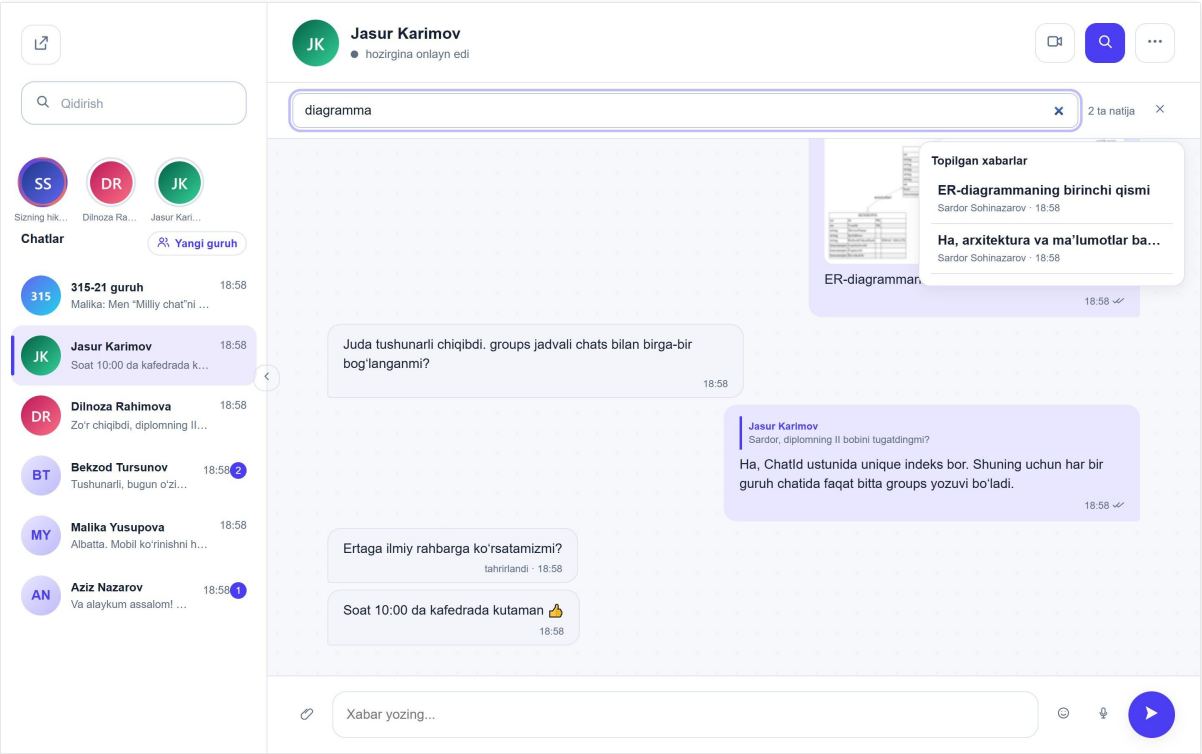
Xabar va fayllar yuboradi, guruhdan chiqadi



Qidiruv, hikoyalar va profil

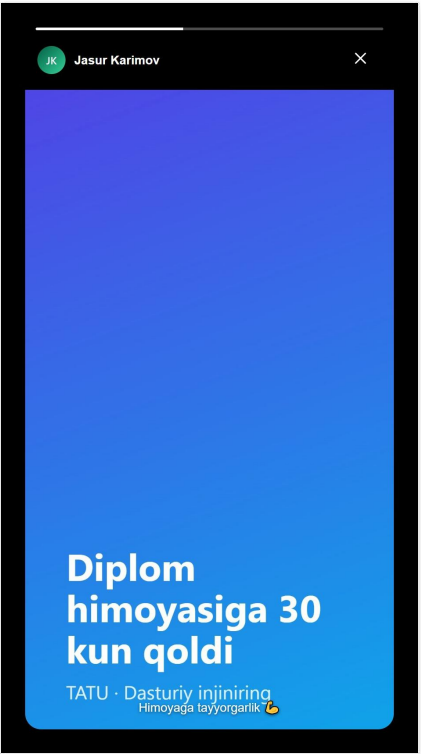
Xabarlarni qidirish

Matn bo'yicha qidirish va topilgan xabar kontekstiga o'tish



Hikoyalar

24 soat ko'rinadi, kim ko'rgani ma'lum



Profil

Ism, username, bio, rasm va faol qurilmalar

Rasmni yuklash
JPG, PNG yoki WebP

Ism

Sardor

Familiya

Sohinazarov

Username

sardor_s

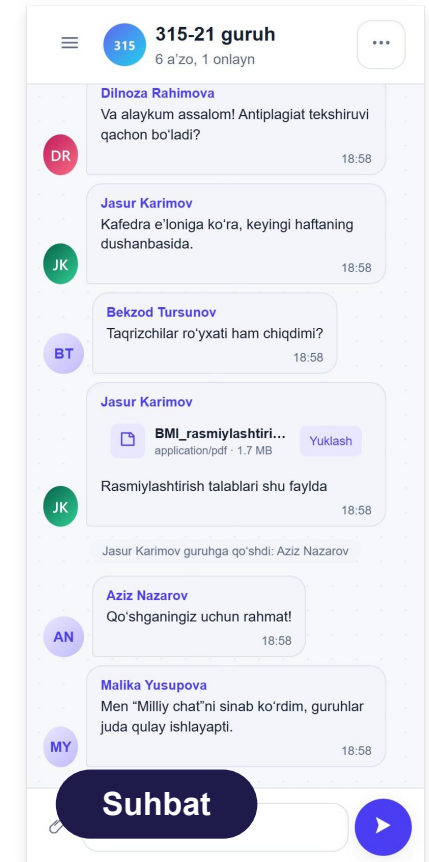
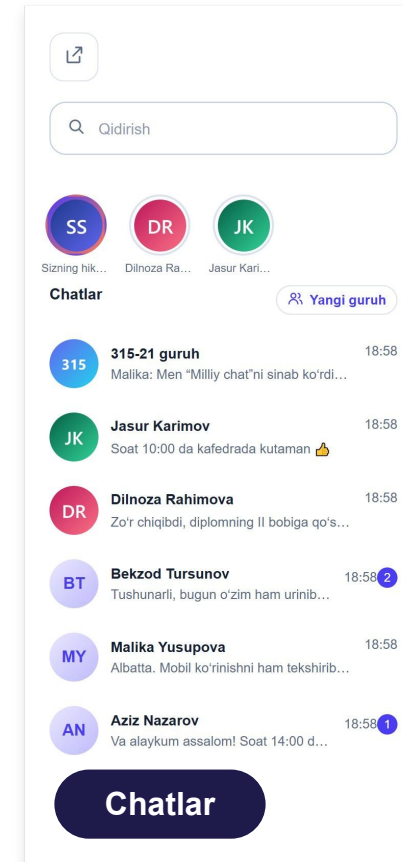
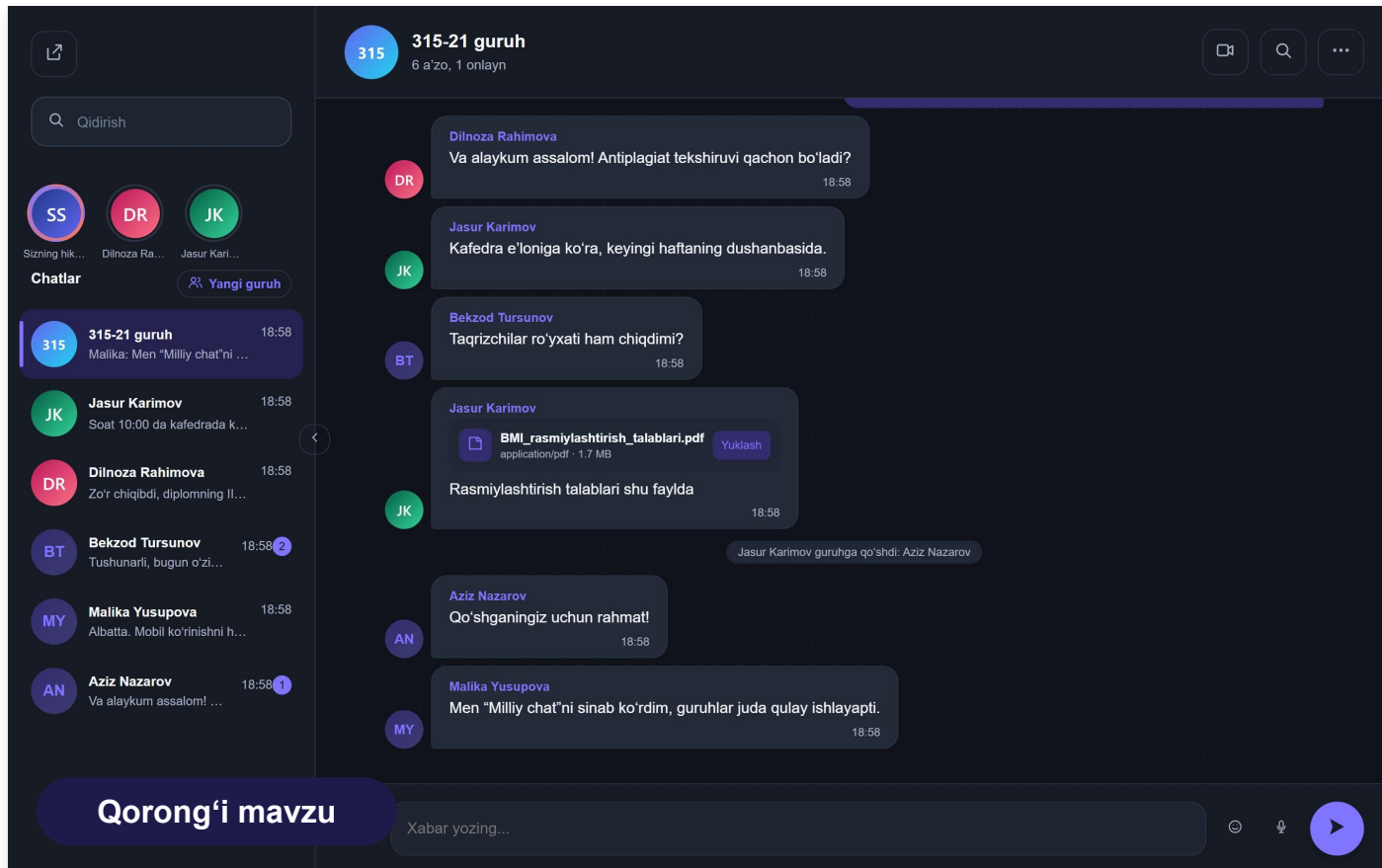
O'zingiz haqingizda

TATU, Dasturiy injiniring. "Milliy chat" muallifi.

sardor.sohinazarov@example.com

Bekor qilish Saqlash

Qorong'i mavzu va mobil ko'rinish



 Bitta kod bazasi kompyuter, planshet va telefonda ishlaydi — o'z dizayn tizimi va moslashuvchan maket.

Joylashtirish va testlash

56

REST API endpointi

39

unit test — barchasi o'tdi

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API integratsion
tekshiruvi

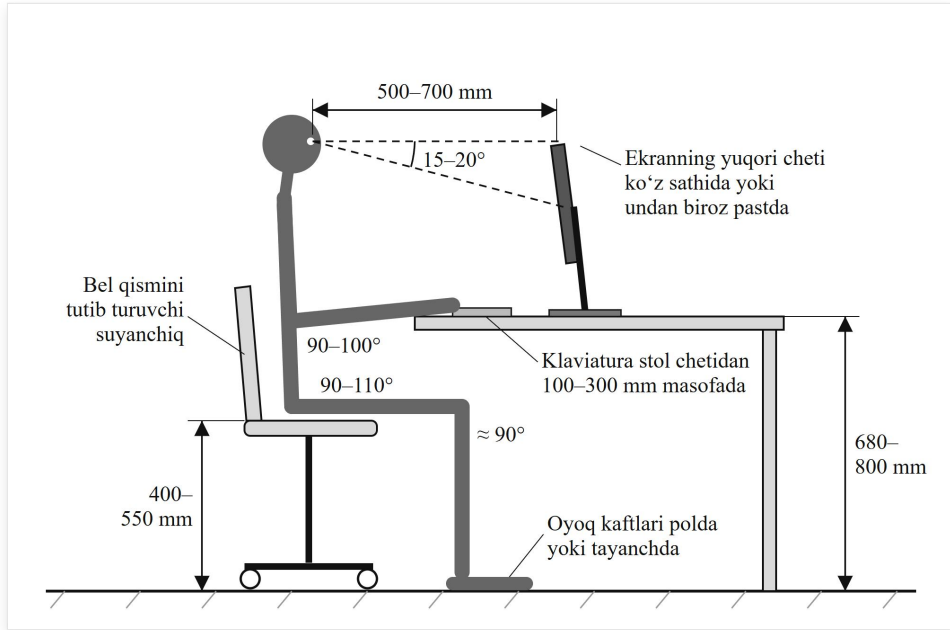
Docker

Render bulutida ishlaydi

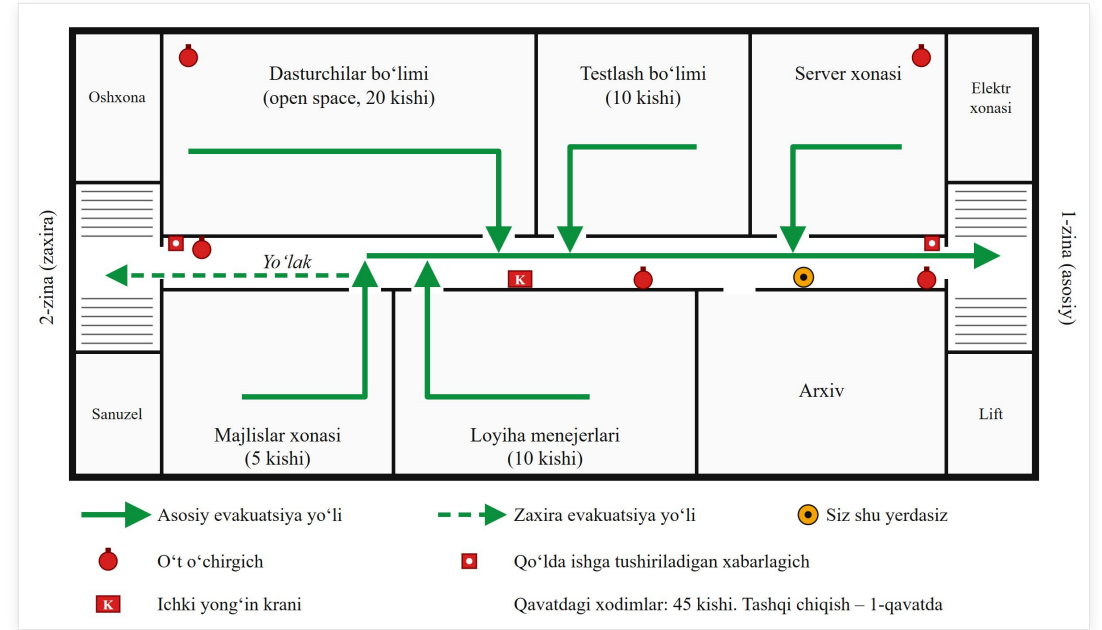
- Ko'p bosqichli Docker obrazi: SDK'da yig'ish, yengil ASP.NET runtime'da ishga tushirish
- GitHub'ga push qilinganda Render obrazni avtomatik qayta yig'adi
- Maxfiy kalitlar kodda emas, muhit o'zgaruvchilarida
- xUnit + NSubstitute: guruhlar, sahifalash, xavfsizlik va validatsiya testlari
- Swagger UI orqali API hujjatlari va qo'lda sinov

```
PS> dotnet test tests\NationalChat.Tests --logger "console;verbosity=normal"
Test run for C:\Users\shohinazarov_x\Desktop\ViDiplom\ish\NationalChat\tests\NationalChat.Tests\bin\Debug\net8.0\NationalChat.Tests.dll (.NETCoreApp,Version=v8.0)
A total of 1 test files matched the specified pattern.
[xUnit.net 00:00:00.00] xunit.net VSTest Adapter v2.5.3.14060a9e5da (64-bit .NET 8.0.31)
[xUnit.net 00:00:00.00] Discovering: NationalChat.Tests
[xUnit.net 00:00:00.10] Discovered: NationalChat.Tests
[xUnit.net 00:00:00.10] Starting: NationalChat.Tests
Passed Security.Pbkdf2OneTimeCodeHasherTests.Verify_MalformedHash_ReturnsFalse [3 ms]
Passed Security.HmacRefreshTokenHasherTests.Hash_IsDeterministicForSameKey [2 ms]
Passed Security.HmacRefreshTokenHasherTests.Constructor_ShortKey_Throws [1 ms]
Passed Security.HmacRefreshTokenHasherTests.Hash_DependentOnKey [1 ms]
Passed Validation.ValidatorTests.UpdateRole_OnlyMemberOrAdmin(role: Admin, expected: True) [15 ms]
Passed Validation.ValidatorTests.UpdateRole_OnlyMemberOrAdmin(role: Member, expected: True) [1 ms]
Passed Validation.ValidatorTests.UpdateRole_OnlyMemberOrAdmin(role: Creator, expected: False) [8 ms]
Passed Validation.ValidatorTests.AddMembers_NonPositiveLimit_IsInvalid [26 ms]
Passed Validation.ValidatorTests.CursorPagination_LimitMustBeBetweenAnd100(limit: 1, expected: True) [10 ms]
Passed Validation.ValidatorTests.CursorPagination_LimitMustBeBetweenAnd100(limit: 101, expected: False) [1 ms]
Passed Validation.ValidatorTests.CursorPagination_LimitMustBeBetweenAnd100(limit: 0, expected: False) [1 ms]
Passed Validation.ValidatorTests.CursorPagination_LimitMustBeBetweenAnd100(limit: 100, expected: True) [1 ms]
Passed Validation.ValidatorTests.CreatedGroup_TitleLongerThanLimit_IsInvalid [1 ms]
Passed Validation.ValidatorTests.CreatedGroup_WithoutMembers_IsInvalid [1 ms]
Passed Chats.ChatPermissionTests.Delete_ByCreatorOrPrivateParticipant_IsAllowed(type: Private, role: Member) [71 ms]
Passed Chats.ChatPermissionTests.Delete_ByCreatorOrPrivateParticipant_IsAllowed(type: Group, role: Creator) [1 ms]
Passed Groups.GroupServiceTests.Get_ByIdMember_ReturnsNull [76 ms]
Passed Chats.ChatPermissionTests.DeleteGroup_ByRoleCreator_IsForbidden [2 ms]
Passed Groups.GroupServiceTests.UpdateRole_OnlyCreatorCanPromote [5 ms]
Passed Groups.GroupServiceTests.Create_EmptyTitle_FailsValidation [2 ms]
Passed Chats.ChatPermissionTests.ClearGroupHistory_ByRegularMember_IsDenied [7 ms]
Passed Groups.GroupServiceTests.AddMembers_ByAdmin_AddsMemberAndNotifiesServiceMessage [9 ms]
Passed Groups.GroupServiceTests.RemoveMember_ByCreator_RemovesAndNotifiesIfNotMovedUser [4 ms]
Passed Groups.GroupServiceTests.AddMembers_OverMemberLimit_IsRejected [1 ms]
Passed Groups.GroupServiceTests.Leave_Creator_TransfersOwnershipToOtherAdmin [2 ms]
Passed Groups.GroupServiceTests.Create_ValidRequest_CreatesGroupWithCreatorRoleAndServiceMessage [4 ms]
Passed Groups.GroupServiceTests.Update_TitleChange_WritesTitleChangeAndServiceMessage [1 ms]
Passed Groups.GroupServiceTests.Leave_LastMember_DeletesChat [1 ms]
Passed Groups.GroupServiceTests.UpdateRole_CreatorRoleCannotBeGranted [1 ms]
Passed Groups.GroupServiceTests.Create_UnknownMember_IsRejected [1 ms]
Passed Groups.GroupServiceTests.AddMembers_AlreadyInGroup_IsRejected [1 ms]
Passed Groups.GroupServiceTests.AddMembers_ByRegularMember_IsForbidden [1 ms]
Passed Groups.GroupServiceTests.RemoveMember_AdminCannotRemoveCreator [1 ms]
Passed Security.Pbkdf2OneTimeCodeHasherTests.Verify_WrongCode_ReturnsFalse [233 ms]
[xUnit.net 00:00:00.84] Finished: NationalChat.Tests
Passed Pagination.CursorPaginationTests.LastPage_ReturnsNoCursor [804 ms]
Passed Pagination.CursorPaginationTests.FirstPage_ReturnsNewestItemsAndCursor [28 ms]
Passed Pagination.CursorPaginationTests.BeforeId_ReturnsOlderItemsOnly [12 ms]
Passed Security.Pbkdf2OneTimeCodeHasherTests.Verify_CorrectCode_ReturnsTrue [228 ms]
Passed Security.Pbkdf2OneTimeCodeHasherTests.Hash_SameCodeTwice_UsesDifferentSaltAndIterationPrefix [230 ms]
Test Run Successful.
Total tests: 39
Passed: 39
Total time: 1.2569 Seconds
```

Hayot faoliyati xavfsizligi



Ergonomik ish joyi va to'g'ri gavda holati



IT-ofis qavatining evakuatsiya reja-sxemasi



5 ish joyi

zona 6 × 6 × 3 m



421 Ik

12 ta LED panel (me'yor
400 lk)



150 m³/soat

toza havo, 5 xodim
uchun



≈ 50 soniya

hisobiy evakuatsiya vaqti

Xulosa va istiqbollar

Erishilgan natijalar

- ✓ Messenjerlar va real vaqt texnologiyalari tahlil qilindi, SignalR tanlandi
- ✓ Clean Architecture asosidagi server, 32 jadvalli PostgreSQL bazasi
- ✓ Parolsiz autentifikatsiya: e-pochta kodi, Google, JWT, sessiyalar
- ✓ Shaxsiy va guruh chatlari, media fayllar, hikoyalar — real vaqtda
- ✓ Angular 22 da moslashuvchan, ikki mavzuli o'zbekcha interfeys
- ✓ Docker orqali bulutga joylashtirildi, testlar bilan tasdiqlandi

Istiqbollar

-  WebRTC ovozli va video qo'ng'iroqlar
-  Uchdan-uchgacha shifrlash (E2E)
-  Kanallar va botlar platformasi
-  Push-bildirishnomalar
-  Android va iOS ilovalari
-  Redis bilan gorizontal kengaytirish



E'tiboringiz uchun rahmat!

“Milliy chat” dasturini ishlab chiqish

S.O. Sohinazarov

Toshkent – 2026